

June 19, 2017

Docket No.: 50-321

NL-17-1114

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D. C. 20555-0001

Edwin I. Hatch Nuclear Plant Unit 1
LER 2017-006-00
Manual Scram Initiated Due to Inoperable Intermediate Range Monitors

Ladies and Gentlemen:

In accordance with the requirements of 10 CFR 50.73(a)(2)(iv)(A), Southern Nuclear Operating Company (SNC) hereby submits the enclosed Licensee Event Report.

This letter contains no NRC commitments. If you have any questions, please contact Greg Johnson at (912) 537-5874.

Respectfully submitted,



David R. Vineyard
Vice President – Hatch

DRV/jcb

Enclosure: LER 2017-006-00

U. S. Nuclear Regulatory Commission
NL-17-1114
Page 2

Cc: Regional Administrator, Region II
NRR Project Manager – Hatch
Senior Resident Inspector – Hatch
RTYPE: CHA02.004

Edwin I. Hatch Nuclear Plant Unit 1

LER 2017-006-00

Manual Scram Initiated Due to Inoperable Intermediate Range Monitors



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOF-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME

Edwin I. Hatch Nuclear Plant Unit 1

2. DOCKET NUMBER

05000321

3. PAGE

1 OF 3

4. TITLE

Manual Scram Initiated Due to Inoperable Intermediate Range Monitors

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
04	20	2017	2017	- 006	- 00	6	19	2017	FACILITY NAME	DOCKET NUMBER
9. OPERATING MODE 4			11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)							
			☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)	
			☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)	
			☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)	
10. POWER LEVEL 0			☐ 20.2203(a)(2)(i)		☐ 50.36(c)(1)(i)(A)		☒ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)	
			☐ 20.2203(a)(2)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(v)(A)		☐ 73.71(a)(4)	
			☐ 20.2203(a)(2)(iii)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(v)(B)		☐ 73.71(a)(5)	
			☐ 20.2203(a)(2)(iv)		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(1)	
			☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(i)	
			☐ 20.2203(a)(2)(vi)		☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(ii)	
			☐ 50.73(a)(2)(i)(C)		☐ OTHER		Specify in Abstract below or in NRC Form 366A			

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT

Edwin I. Hatch/Jimmy Collins – Licensing Supervisor

TELEPHONE NUMBER (Include Area Code)

912-537-2342

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX

14. SUPPLEMENTAL REPORT EXPECTED

☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE)☒ NO

15. EXPECTED SUBMISSION DATE

MONTH DAY YEAR

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On April 20, 2017 at 2345 EST with Unit 1 in Mode 4 at 0 percent rated thermal power, the Unit 1 Reactor Mode Switch was taken from the REFUEL to the SHUTDOWN position per Technical Specification (TS) Limiting Condition for Operation (LCO) 3.10.4, Required Action A.2.2, due to having no operable intermediate range monitors (IRMs) in one quadrant of the reactor vessel because of maintenance activities. All control rods had been previously inserted, therefore no control rod motion occurred when the mode switch was positioned to SHUTDOWN.

The cause of this event was due to flawed technical decision making surrounding maintenance activities related to inoperable IRMs. An inadequate review of TS LCO 3.10.4, in part due to the emergent nature of the maintenance activities to restore IRMs to operable status, resulted in Operations' incorrect determination that one of the requirements of TS LCO 3.10.4, "Single Control Rod Withdrawal – Cold Shutdown" was not met. The Unit 1 Reactor Mode Switch was therefore placed to the SHUTDOWN position per TS 3.10.4, Condition A.2.2.

Corrective actions included ensuring TS Bases 3.10.4 language is not ambiguous, issuing an order that requires the shift manager to approve the removal of TS equipment from service for maintenance, and inclusion of this event in licensed operator continuing training.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO
Edwin I. Hatch Nuclear Plant Unit 1	05000-321	2017	- 006	- 00

NARRATIVE**Event Description**

On April 20, 2017 at 2345 EST with Unit 1 in Mode 4 at 0 percent rated thermal power, the Unit 1 Reactor Mode Switch was taken from the REFUEL to the SHUTDOWN position per Technical Specification (TS) Limiting Condition for Operation (LCO) 3.10.4, Required Action A.2.2, due to having no operable intermediate range monitors (IRMs) in one quadrant of the reactor vessel because of maintenance activities. All control rods had been previously inserted, therefore no control rod motion occurred when the mode switch was positioned to SHUTDOWN.

Event Cause Analysis

The cause of this event was due to flawed technical decision making surrounding maintenance activities related to inoperable IRMs. An inadequate review of TS LCO 3.10.4, in part due to the emergent nature of the maintenance activities to restore IRMs to operable status, resulted in Operations' incorrect determination that one of the requirements of TS LCO 3.10.4, "Single Control Rod Withdrawal – Cold Shutdown" was not met. The requirements of TS LCO 3.10.4 dictate that the MODE 5 requirements of TS LCO 3.3.1.1 for operable IRMs be met. In MODE 5, with a control rod withdrawn in a cell containing fuel, TS LCO 3.3.1.1 requires that there be an operable IRM in each quadrant. In this case, since all rods were already inserted, LCO 3.3.1.1 MODE 5 requirements for IRMs were not applicable, thus the LCO 3.10.4 requirements for IRMs were met. However, due to the incorrect determination that this requirement of TS LCO 3.10.4 was not met, the Unit 1 Reactor Mode Switch was placed to the SHUTDOWN position per TS 3.10.4, Condition A.2.2.

Safety Assessment

This event is reportable per 10 CFR 50.73(a)(2)(iv)(A) as an event that resulted in the automatic actuation of the reactor protection system (RPS).

The IRMs monitor neutron flux from the upper portion of the source range monitor (SRM) range to the lower portion of the power range neutron monitoring (PRNM) range. The IRM system consists of eight IRM channels and detectors that are inserted into the core for a reactor startup and are withdrawn after the reactor mode selector switch is positioned to RUN. The IRM system is divided into two groups of IRM channels arranged in the core. IRM channels "A", "C", "E", and "G" are associated with trip system A of the RPS and IRM channels "B", "D", "F", and "H" are associated with trip system B. There are two IRM channels in each quadrant of the core. The IRM system is capable of generating a trip signal that can be used to prevent fuel damage resulting from abnormal operating transients that occur while operating in the intermediate power range.

There was no change in reactivity and no movement of the control rods occurred since all rods were inserted prior to this event. Therefore, this event is considered to have low safety significance.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO
Edwin I. Hatch Nuclear Plant Unit 1	05000-321	2017	- 006	- 00

Corrective Actions

The TS Bases were reviewed to ensure there was no ambiguity in the wording of TS Bases 3.10.4. A Standing Order was issued to provide direction that the Shift Manager must approve any work that removes TS equipment from service. Additionally, this event will be covered in Licensed Operator Continuing Training.

Previous Similar Events

None.